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Leveraging Modality-Agnostic Self-Distillation Transformers for Multimodal Sarcasm Detection

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Leveraging Modality-Agnostic Self-Distillation Transformers as Feature Extractors for Multimodal Sarcasm Detection

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This article investigates the application of self-distillation transformers as feature extractors for multiple modalities in the task of multimodal sarcasm detection. Specifically, we choose Data2Vec to extract features from the three channels of data from the high-quality multimodal dataset in sarcasm detection, MUStARD++. Using these features, we first examine their effectiveness in unimodal sarcasm detection, then move onto multimodal sarcasm detection, where we conduct ablation studies by withholding certain modalities to the sarcasm detection model. We examine how the features extracted from three separate modalities perform and whether their performance is impacted by Data2Vec's unified cross-modal learning objective and modality-agnostic, contextualized latent representations. Finally, we review all results to determine if sharing a unified learning objective across modalities in a single self-supervised learning framework might impact the robustness of its learned representations in regards to a multimodal NLP task that engages with all the modalities it is designed to process, as well as discuss future research directions.

Keywords: multimodality, sarcasm detection, self-supervised learning, self-distillation

1. Introduction

Multimodality has emerged as a promising direction for many computationally-based research fields including social sciences (O'Halloran, Pal, and Jin 2021), neurobiology(Lotter et al. 2023), robotics (Su et al. 2023), but most importantly computational linguistics (CL) (Parcalabescu, Trost, and Frank 2021) and natural language processing (NLP) (Zhang et al. 2020). As NLP research continues to benefit from the development of increasingly powerful graphics processing units (GPUs), there has been a growing interest in how multimodality could impact past research mostly built on traditional text-based NLP techniques (Bouazizi and Otsuki Ohtsuki 2016), particularly in cases where information in multiple modalities are salient to the research topic (Schifanella et al. 2016). Whether in discourse analysis (O'Halloran 2011), emotion detection (Abdullah et al. 2021), opinion mining (Li et al. 2019), the turn to multimodality signifies a new chapter with its unique promises and challenges.

Particularly in areas like detection of human emotion and sentiment analysis (Das and Singh 2023), multimodal NLP aims to more closely examine the blurred lines between semantics and pragmatics in human communication. Sarcasm detection, under the umbrella task of sentiment analysis, is to correctly identify verbal irony where someone says the opposite of what they truly mean. From a neuro-biological perspective, sarcasm is a high level cognitive function that simultaneously utilizes parts of the brain engaged with sight, sound, feelings, language understanding, and abstract

reasoning. (Shamay-Tsoory, Tomer, and Aharon-Peretz 2005). This complex linguistic phenomenon oftentimes have a social function, displaying an impact on interpersonal relations between friends and strangers (Caucci and Kreuz 2012). Despite its prevalence in day-to-day scenarios, sarcasm can be tremendously difficult to interpret even for humans, as it is often facilitated by an array of paralinguistic cues that are heavily context and cultural dependent (Camp 2012). Such cues include unusual delivery in pace and intonation, exaggerated emphasis on certain words, unorthodox word choice, or even gestures and facial impressions. For instance, "Oh, great" said in a cheery tone might be taken literally, while the same phrase with a rolling of eyes and a stressed tone would be understood as sarcastic. This could be further compounded by the subversion of expectation, as a "correct" response might vary from culture to culture, and even from speech patterns of any given individual to another depending on gender, race, education level...etc.

While introducing multimodal learning to the sarcasm detection process could help computers with accounting for these nontrivial cues, multimodal NLP regardless of early or late fusion (Gadzicki, Khamsehashari, and Zetzsche 2020) relies on robust features extracted with State-of-the-Art (SOTA) models for all modalities it aims to leverage. To train a single, unimodal model that is able to effectively extract features requires tremendous amount of time and human labor in annotation built on previous research, and this demand for high quality feature extraction grows rapidly as an NLP task engages with more and more modalities. One aisle of machine learning that seeks to mitigate this problem is self-supervised learning (SSL). While the exact delineation of SSL is debated by many (Balestriero et al. 2023), a general agreement is it describes as any kind of learning that doesn't require explicit supervising but instead uses structure inherent to the training data as supervisory signal. SSL allows researchers to turn massive amounts of noisy data with little to no curation into effective training corpus through cleverly designed *representation learning* (Zhong et al. 2016) pretext tasks, acquiring descriptive and meaningful abstractions which allow them to perform well on downstream tasks or be used as feature extractors (Balestriero et al. 2023). Within the context of multimodal sarcasm detection in NLP, this opens up a vast repository of potential high quality feature extractors.

However, as aforementioned SSL relies on structure inherent of the data, in a self-supervised multimodal learning (SSML) framework, the blackbox problem of its representations (shared by most deep learning algorithms (Hussain 2019)) is exacerbated by a central question of what defines a modality (Zong, Aodha, and Hospedales 2023). This means when deploying SSL/SSML models as feature extractors in a multimodal NLP task, researchers face a dizzying amount of variables. In the research of multimodal sarcasm detection, application of SSL is most widespread for the text modality, while other modalities often deploy conventional, non-SSL feature extractors. Even in rare cases where SSL feature extractors are deployed on the audio and/or vision modality in a bimodal (Maity et al. 2022) or trimodal setting (Bhosale et al. 2023), the selection of the SSL model is largely dependent on its previous unimodal benchmarking or with the sole objective of improving performance. This has implications for researchers' understanding of SSL, whose works tends to be empirically driven, meaning while these models come with numerous moving pieces that may impact key properties of the final representations, the implications of these variables are not necessarily well-detailed in published work (Balestriero et al. 2023). As such, how the modality-specific learning heuristics of selected SSL/SSML models in a multimodal sarcasm detection impact overall performance is severely underresearched.

Self-distillation (SD), an emerging branch of SSL, shows promise to provide some insight into this issue. Due to the architecture of the student-teacher network in SD, the same pretext tasks for learning representations could be reused across distinct modalities of data (Zong, Aodha, and Hospedales 2023). Data2Vec (Baevski et al. 2022) achieves this by unifying the learning objective of masked token prediction within a student-teacher network for text, vision, and audio, and experiments on key benchmarks in natural language understanding, speech recognition, and image classification all showcase either a new SOTA performance or competitiveness with leading methods.

This paper thus investigates multiple things: 1) to examine the behavior of latent representations learned by the modality-agnostic, self-distillation transformers of Data2Vec by applying it to a multimodal task where all its target modalities are engaged, and 2) to add to the limited research on the application of SSL models in multimodal sarcasm detection. Applying Data2Vec to a multimodal sarcasm detection benchmark MUStARD++, we hypothesize that compared to the previous research, while models using only features from a single modality may perform better or worse depending on finer details specific to that modality, combinations of two or more modalities may give an edge to Data2Vec due to its unified learning objective. In detail, we first extract features with Data2Vec-text, Data2Vec-audio, and Data2Vec-vision from the three channels of input modality from MUStARD++. We then select a variety of different input, tokenization, and feature extraction techniques for later experimentation. Replicating the existing binary sarcasm classification model from the MUStARD++, we conduct pretesting and hyperparameter search. Lastly, we conduct two sets of final ablation studies of modality input, context information, and speaker information collect these results, and compare them to previous results on this dataset.

To lay the foundation to this paper, we posit the following questions:

1. How do SSL/SSML models proven to be highly effective in various benchmarks perform as feature extractors in the task of multimodal sarcasm detection?

2. In the task of multimodal sarcasm detection, are there notable dynamics between different modality combinations that can be attributed to the learning heuristics of the SSL/SSML feature extractors used?

The paper structure is as follows: in Section 1 we offer an introduction and pose the research questions regarding the general applicability of SSL models as feature extractors for the downstream task of multimodal sarcasm detection. In Section 2, we discuss past academic research conducted in the broader field of multimodal sarcasm detection. We also review a few prominent SSL/SSML learning heuristics, establish their use case, and address their shortcomings. In Section 3, we elaborate on our methodology by introducing the dataset we use, providing relevant theoretical details of Data2Vec, and describing our sarcasm detection model and ablation studies. In section 4, we present the results to our experimentation and discuss the implications of our findings. In Section 5, we draw conclusion by acknowledging the limitations of this paper and propose future research directions.

2. Related Works

2.1 Multimodal Sarcasm Detection

Multimodal sarcasm detection as a practical subject expands across multiple disciplines and use cases. In e-commerce, designing artificial intelligence or recommendation systems that are able to interpret veiled dissatisfaction and accurately summarize the sentiment of reviews can both inform potential customers and provide relevant feed-

back to service providers to aid them in adjusting their services (Erfina et al. 2020). In online communities, sarcasm detection may help with monitoring hate speech concealed within some form of verbal irony (Frenda et al. 2018). In medical care, sarcasm detection could help identify speech patterns that correlate to early signs of anxiety and depression (Rothermich, Ogunlana, and Jaworska 2021), or serve as an auxiliary tool to measure mental decline in Alzheimer's patients who struggle to understand rhetoric devices like metaphor and sarcasm (Maki et al. 2013). In social sciences, discourse analysis and topic modelling heavily rely on effectively interpreting debate, such as in a political setting where scathing and pointed sarcastic comments likely disproportionately circulate more than straightforward remarks (Kannangara 2018).

The key caveat of multimodal sarcasm research is that the inputs of modalities could be incredibly diverse depending on the use case in which said sarcasm occurs. Researches range from working with commonly accessible modalities of data, such as text-vision pairs (Xu, Zeng, and Mao 2020), text-audio pairs (Michael and Zahra 2019), and audio-vision pairs (Tomar et al. 2023); to working with data that requires a fairly complex and costly setup to acquire, which may not be used outside a laboratory setting, such as text-gaze feature pairs (Mishra, Kanojia, and Bhattacharyya 2016; Mishra, Dey, and Bhattacharyya 2017) and text-electroencephalogram pairs (Filik et al. 2014). For the sake of this study, we will only focus on multimodal sarcasm detection using the three modalities most commonly accessible for NLP research : text, vision, audio.

2.1.1 MUStARD and Sarcasm Detection in Conversational Context. While the role of modality has been a subject of interest for NLP researchers investigating sarcasm detection, high quality and substantially sized multimodal datasets that have an emphasis on sarcastic utterances are hard to come by. In 2019, the Multimodal Sarcasm Detection Dataset (MUStARD) (Castro et al. 2019) paper is published where the researchers constructed and annotated a multimedia collection of 6365 short video clips, 345 of which contain sarcastic utterances and 6020 do not, and the three modalities it contains are text (transcripts of speech), audio and video (naturally aligned modalities). Along with this, the researchers identify a novel sub-task in multimodal sarcasm detection, *sarcasm detection in conversational context*. Video clips in the positive sarcasm class are collected from the TV shows Friends, The Golden Girls, Big Bang Theory (seasons 1–8), and Sarcasmaholics Anonymous, while video clips in the negative sarcasm class are selected from the Multimodal EmotionLines Dataset (MELD) dataset (Poria et al. 2019). A rigorous inter-annotator agreement procedure produces a Kappa score of 0.5877. In addition to the sarcastic utterances themselves, the authors of MUStARD also add context information in the form of preceding speech of people exchanging a few sentence that is immediately followed by the utterance in question. Along with this, the speaker of the sarcastic utterance are also labelled to further round out the dataset.

The authors then design a multimodal binary sarcasm classification approach by first extracting features from the three modalities separately: using BERT-base encoder (Devlin et al. 2019) for the text-modality, classical audio processing low-level features (i.e. MFCC, Mel-Spectogram, spectral centoid, associated temporal derivatives) for the audio modality (McFee et al. 2015), and the pooling layer of Resnet-152 (He et al. 2015) averaged across the entire video for the visual modality. Performing modality fusion before training and testing a Support Vector Machine (SVM) (Cortes and Vapnik 1995) model across a 5-fold cross-validation ablation studies experiment, researchers achieve an error rate reduction of up to 12.9 percent in certain ablation configurations compared to unimodal models trained on the same dataset (F1-score of 63.1% when using the text and audio modality).

Some subsequent researches focus on improving the prediction performance on the dataset by optimizing selected channels of input modalities. Bharti et al. (2022) believes the visual modality to not contribute much to the detection of sarcasm in MUStARD, and thus experiments with only combining text input and audio input but exploring a wide variety of classifiers. By experimenting with embeddings created by Word2Vec (Mikolov et al. 2013), ELMo (Peters et al. 2018), and BERT (Devlin et al. 2019), and a custom RNN-based audio feature extractor build on top of MFCC representations, they achieve a new high in F1-score of 69% with a SVM model. In Helal et al. (2024), researchers hone in on the singular modality of text and experiment with finetuning RoBERTA (Liu et al. 2019) and DistilBERT(Sanh et al. 2020) on the News Headline dataset (Misra and Arora 2023), then using the finetuned models to create embeddings. The resulting embeddings in their classification experiments achieve a stunning F1-score of 90%.

Other researchers focus on enriching the MUStARD dataset with additional labels to improve sarcasm detection performance. In Chauhan et al. (2020), researchers hypothesize that an incongruity between surface level emotions, that are outwardly displayed, and hidden emotion, inwardly harbored by the speaker based on context, can aid in the prediction of sarcasm Chauhan et al. (2020). By annotating the original MUStARD dataset with explicit emotions and implicit emotions of 9 different classes, the researchers are able to improve the performance of sarcasm detection using these labels during prediction with a F1-score of 72.6%. In Chauhan et al. (2022), researchers first annotate each utterance within the original MUStARD dataset with a corresponding emoji, the emoji's sentiment, and the emoji's emotion. Following this they propose an emoji-powered-multimodal deep learning framework with multimodal sarcasm detection as its primary task and sentiment/emotion detection in a conversational scenario its secondary task, ultimately reaching a F1-score of 77.9% for the sarcasm detection task. In Tomar et al. (2024), researchers leverage previous research done on dialogue acts (Stolcke et al. 2000) and annotate the MUStARD dataset with Dialogue Acts tags (e.g. Greeting, Question, Answer, Statement-Opinion, Statement-Non-Opinion...etc). Training a novel modality fusion classification model with text embeddings from BART (Lewis et al. 2019), audio embeddings from Wav2Vec2 (Schneider et al. 2019), and visual embeddings from Resnet-152 (He et al. 2015), they find that dialogue act-aided sarcasm identification achieved better performance compared to performing sarcasm identification alone, with a F1-score of 81.2%.

Yet other researchers hone in on specifically modelling the multimodality aspect of sarcasm in a more fine-grained manner. In Wu et al. (2021), researchers propose the incongruity-aware attention network (IWAN), which improves sarcasm detection by creating a scoring mechanism that aims to model on the word-level specific incongruity between modalities. In M2Seq2Seq Zhang et al. (2023) introduces a dual attention framework where two attention mechanisms are designed separately to either capture the contextual dependency between adjacent utterances within the same modality (intermodal attention), capture the interactions between two different modality inputs. In Pramanick, Roy, and Patel (2022), researchers similarly focus on modelling intramodal and intermodal dynamics of sarcasm but use optimal transport (Peyré, Cuturi et al. 2019) to model intermodel dynamics instead of attention mechanism.

Despite being a well researched dataset, some (Ray et al. 2022; Bhosale et al. 2023) highlight the small size of sarcastic utterances and lack of diversity of the MUStARD dataset, which contribute to two potential problems: tendency of models to overfit and fail to later generalize on other multimodal sarcasm detection works, and sensitivity that contributes to dramatic improvements or drops in performance. As such, these

researchers curate expanded version of the dataset focusing on diversifying and increasing the number of video clips, as opposed to adding new annotation labels to existing clips, creating the MUStARD++ (Ray et al. 2022) and the MUStARD++ Balanced (Bhosale et al. 2023) datasets.

2.2 Self-supervision within Multimodal Learning

According to Goodfellow (2016), self-supervised learning (SSL) is a form of unsupervised learning where the data itself provides the supervision. Rather than relying on labor-intensive datasets labelled with ground truth or prediction targets to use as supervisory signal, self-supervised models are designed to create implicit *labels* from unstructured data that can serve as supervision. This usually involves label-free pretext tasks (i.e. pretraining objective that is tailored to a modality of data and poses a challenge for the model to overcome) like instance discrimination (i.e. contrastive learning) or masked prediction objectives (2.2.1). A prime example that demonstrates the effectiveness of SSL's representation learning is the large language model (LLM), which has seen great success in text-based natural language processing tasks (Liu et al. 2019; Devlin et al. 2019; Liu et al. 2023). The transferring of this learning strategy to other modalities beyond just text has received much attention, as being able to extract features from other modalities as effectively as LLMs from text could facilitate computers to interpret human communication multimodally (Zong, Aodha, and Hospedales 2023).

However, the introduction of SSL to a multimodal learning setting makes self-supervised multimodal learning (SSML) a loaded term. In Zong, Aodha, and Hospedales (2023), researchers identify multiple scenarios where the terminology "multimodal SSL" is used:

1. Learning without labels from automatically paired multimodal data, such as movie clips with naturally synchronized vision and audio data or images paired with depth data from RGBD (Red, Green, Blue, Depth) cameras.
2. Learning from paired multimodal data which was originally created for a different purpose, e.g. CLIP (Radford et al. 2021) with image-caption pairs scraped from the web or ViFiCLIP (Rasheed et al. 2023) with video clip-caption pairs. As these paired modalities are not created specifically with the supervision goal in mind, some researchers consider this a case of multimodal SSL learning in practicality.
3. Learning that uses both free and manually annotated multimodal data at large scale, which aligns with SSL's spirit of "scale up by breaking the manual annotation bottleneck" (Zong, Aodha, and Hospedales 2023)

Without overanalyzing the exact definition of SSL, for the scope of this paper, we will only review SSL/SSML whose works or mechanisms are relevant to modalities commonly used in multimodal NLP. In brief, we describe two families of SSL/SSML and their demonstrated success in a variety unimodality benchmarks.

2.2.1 Masked Token Prediction. One of the most well developed and versatile approach to self-supervised learning is the task of Masked Token Prediction (MTP). An early prominent example of this would be BERT (Devlin et al. 2019), where researchers proposed Masked Language Modelling (MLM) as one of the two pretext tasks that allow

for BERT to gain understanding of the meaning of language inherent to texts without positive or negative labels. By randomly masking out text tokens and prompting the model to predict the missing information, high dimensional representations are learned when pretraining BERT on a large corpus of BooksCorpus (Zhu et al. 2015), which contains 800M words, and English Wikipedia, which contains 2,500M words. These learned representations prove to be useful even for downstream tasks not directly related to the initial pretext learning objective (e.g. sentiment analysis, named entity recognition...etc), being able to improve the performance of an array of natural language processing tasks with fine-tuning, or in certain cases even without (Devlin et al. 2019). This prompted a wave of large scale pretrained NLP models working with discrete tokens who utilize the same training objective, such as RoBERTa and CamemBERT (Liu et al. 2019; Martin et al. 2020), or even not strictly NLP protein sequence models who similarly work with discrete tokens like ProteinBERT (Brandes et al. 2022).

Outside of NLP, masked language modelling as a learning strategy for self-supervised learning is also successful when transferred to new modalities of data in other fields, such as audio signal processing and computer vision, in the form of *masked token prediction*. For audio processing, Wav2Vec2 (Schneider et al. 2019) tackles three distinct challenges that is specific to audio processing: (1) each input utterance contains multiple sound units, (2) no predefined lexicon of input sound units during pre-training, and (3) sound units vary in length without clear segmentation. As such, researchers devise a novel masking strategy (section 3.1.3) of spans of audio and encouraging the model to retrieve missing information in a MTP task. Similarly, HuBERT (Hsu et al. 2021) also uses the masking approach but adds a second learning objective of classification task, forcing the model to classify hidden sequences into predefined categories. In computer vision, MTP's proven success comes in the form of Bidirectional Encoder representation from Image Transformers (BEiT) (Bao et al. 2022), where researchers build upon the success of Vision Transformer (ViT) (Dosovitskiy et al. 2021) and mask images randomly in a MTP task by creating 16x16 pixel patches then making the model recover the missing patches, achieving a 83.2% top-1 accuracy on ImageNet-1K (Deng et al. 2009).

2.2.2 Self-Distillation. The family of self-supervised learning methods known as Self-Distillation (SD) methods are inspired by a different group of SSL methods, contrastive learning or Deep Metric Learning (DML) methods (Balestriero et al. 2023). In brief, DML trains a model to predict whether two inputs belong to the same class by making their embeddings close to each other, or to different classes by making their embeddings far apart. In the context of SSL, since the data lacks labels, variations of the original input data is augmented using known *semantic-preserving transformations* (e.g. color jittering, image cropping, image rotation...etc in the case of computer vision) to create similar inputs that can be considered the same class. These variations that contain the same subject are referred to as positive pairs, while the samples that models aim to distinguish are called negatives. With the learning objective of reducing contrastive loss between the output of the positive pairs, useful representations are learned that can be used for downstream tasks.

Self-distillation methods builds upon DML, and addresses the main problem of *representation collapse*. *Representation collapse* is when a model fails to penalize similar representations for dissimilar samples effectively, or overfitting to positive pairs at the expense of considering negative pairs, resulting in embeddings that are too similar to one another. While this achieves the goal of reducing contrastive loss, it defeats the purpose of contrastive learning, as no meaningful representation ends up being learned.

In grave cases, total collapse of representations may occur, rendering the entire model unable to transfer any degree of learning to downstream tasks. This could happen due to poor curation of positive/negative pairs and batch sizes issues, meaning DML are both sensitive to the exact semantic-preserving transformation used on its training data and batch sizes that are too small to introduce the model to enough negative pairs in a given timestep.

In Grill et al. (2020), researchers present "Boostrap Your Own Latent" (BYOL) and outline an innovative paradigm for self-supervision learning by leveraging knowledge distillation as a means to prevent representation collapse. Traditionally, knowledge distillation (KD) is the process of freezing the weights of one bigger model (Teacher Model) and tuning a second, smaller model (Student Model) by gradually reducing the contrastive lost between the two model. BYOL (Grill et al. 2020) first initiates a target Convolutional Neural Network (CNN), analogous to the Teacher Model in KD, and an online CNN network, analogous to the Student Model in KD. But instead of transfer learning knowledge from the teacher to the student, BYOL experiments with using the same model in both roles by first randomly initiating the weights of the target network, creating representations of a sample with these weights. The online network is then forced to align its predicted representations of an augmented version of the sample to the unaugmented representations of target network. By creating an iterative *delay* between update of weights for the target vs. online network (making the update of weights for the target network the exponential moving average of the online network), this self-distillation framework manages to avoid representation collapse while learning some features despite the making a student learn from representations generated by a randomly initialized network. By further asymmetrizing the target-online network, combined with carefully selected regularization, loss function, and appropriate data augmentation methods used in prior DML research, BYOL achieve SOTA results that closely match fully-supervised vision models (Grill et al. 2020) without the downsides of DML, as it is much less reliant on the introduction of numerous negative pairs in comparison.

Later self-distillation works like the computer vision model DINO (Caron et al. 2021) explore upon this self-distillation framework and prove that Vision Transformers also adapt well to this learning heuristic. A later iteration, DINOv2 (Oquab et al. 2024), further integrates the Masked Token Prediction learning objective into a student-teacher network by forcing a student model that has patches of images masked to predict the representations of a teacher model.

3. Research Methodology

This study aims to examine how do SSL models, more specifically the three modality-agnostic self-distillation transformers of Data2Vec, perform as feature extractors for the task of binary multimodal sarcasm detection on the MUStARD++ dataset. Additionally we investigate whether the *contextualized latent representations* generated by Data2Vec-text, Data2Vec-vision, and Data2Vec-audio exhibit special cross-modal interactions due to their unified pretext learning objective, despite them being pretrained on separate modalities of data.

We hypothesize that despite limited past studies on usefulness of SSL feature extractors for multimodal sarcasm detection, using Data2vec on all three modalities could prove to be performant as Data2Vec-base models from each individual modality have reached excellent results on multiple unimodal benchmarks (Baevski et al. 2022). We also hypothesize that because of the unified masked token prediction and self-

distillation framework of Data2Vec across modalities, not only would we witness a consistent increase in performance as more modalities are used as input, but also changes in performance when specific modalities are used together or left out as input that reflect the modality-agnostic pretext task of predicting contextualized latent representations.

To test out these hypotheses, we extract features with Data2Vec-text, Data2Vec-audio, and Data2Vec-vision from the MUStARD++ dataset. We then use these to train binary sarcasm classification model and conduct ablation studies by selectively withholding modality and evaluating their performances. In all experiments, we use the same implementation of Data2Vec hosted by Huggingface Transformers. While we acknowledge that a newer version Data2Vec 2.0 (Baevski et al. 2023) exists, the purpose of this paper is to observe in principality how the representations of modality-agnostic SSL transformers perform in a multimodal NLP task setting where all its target modalities are engaged. As there are no fundamental changes between the architecture of Data2Vec and Data2Vec 2.0 that would affect this shared aspect of the two, we choose Data2Vec as our feature extractor due to the rich literature surrounding it.

We first highlight the architecture, learning mechanism, and performance of Data2Vec in section 3.1, then provide details to the MUStARD++ dataset and its experiments in section 3.2. In section 3.3 we describe preliminary data cleaning of the dataset, and in section 3.4 we elaborate on the preprocessing, tokenization, and feature extraction process with Data2Vec of all three modalities of the cleaned MUStARD++. In section 3.6 we explain the architecture of the binary sarcasm classifcation model used, and in section 3.6.2 we detail a series of pretesting we conduct for hyperparameter search and exploratory purposes. Finally, in section 3.6.3, we provide the details to our final sarcasm detection ablation studies.

3.1 Data2Vec

In (Baevski et al. 2022), Meta AI proposes Data2vec, a new self-supervised learning framework that is designed to unify the objective of representation learning across three different data modalities: text, speech, and vision. Data2Vec builds upon the success of an array of successful unimodal self-supervised learning techniques, combining the Masked Target Modeling tasks as seen in BERT (Devlin et al. 2019) and Wav2Vec2 (Baevski et al. 2020) (section 2.2.1), with the self-distillation strategy seen in BYOL (Grill et al. 2020) and DINO (Caron et al. 2021) (section 2.2.2), to create a consistent learning heuristic that can be applied across various modalities of input data.

The central concept behind Data2Vec is to predict the latent *contextualized representations* of complete input data from a masked view within a self-distillation framework, instead of any type of modality-specific target. Utilizing a standard Transformer architecture to accomplish this task, the self-distillation framework concentrates on these contextualized representations, which are the internal layers of a neural network. As Data2Vec operates without needing to predict specific visual tokens, words, or sounds, lack of necessity for modality-specific targets during the learning process also means a single algorithm can work effectively with different kinds of inputs.

Like other self distillation models, Data2Vec begins by creating two modes to the same model: Teacher mode and Student mode. In Teacher mode, the backbone model (specific to the data type) generates embeddings for the entire input data to serve as targets, akin to the case of creating pseudolabels. In Student mode, the same backbone model creates embeddings for a masked version of the input data, attempting to predict the complete data representations made by the teacher mode.

- In Teacher mode, the target is a weighted sum of all features within a sample. For instance, with image data, the backbone model generates representations of four distinct patches of the image and computes a weighted average for each. The training targets are obtained by averaging the top K Feed-Forward module blocks from the teacher encoder where *K* is a hyperparameter. We discuss further implications about top K block selection in section 3.5.
- In Student mode, parts of the input data are masked. The student encoder network generates a representation for only the unmasked input data. The model then inserts constant values in place of the masked data patches. This resultant embedding passes through the student decoder network, which tries to predict the target representation created by the teacher encoder network. The learning process compares only the masked input representation between the teacher and student networks.
- The weights of the teacher encoder network update slower than the weights of the student encoder network, being the exponentially moving averages (EMA) of the student encoder weights. To compare the target representation (from the teacher network) with the masked representation (from the student network), a Smooth L1 loss is used to regress these targets.
- In a word, the teacher network and the student network display a dynamic behavior: student's parameters are updated by optimizing the objective function in step 2, while the teacher's parameters are updated by calculating the the EMA in step 3. (This is the same basic mechanism BYOL (Grill et al. 2020) uses to avoid representation collapse)

Two key attributes of Data2Vec embeddings, compared to other SSL/SSML models, are their "continuous" and "contextualized" nature. Continuous embeddings, which are real-valued vectors, offer an advantage over embeddings of discrete tokens by not being limited to a fixed vocabulary size, allowing new words to be embedded seamlessly. Moreover, unlike earlier self-supervised models such as BERT, Data2Vec embeddings are naturally contextualized due to the use of transformers and positional encoding in its structure, meaning they capture information not only about a single token but also about the tokens surrounding it. Authors of Data2Vec believe this incentivizes the model to learn non-local features and attune to more long-range dependencies, makeing the latent representations of Data2Vec inherently more rich compared to a fixed set of target(s) based on local context such as those used in previous self-supervised Masked Token Prediction work.

However, it is important to note that Data2Vec still relies on separate tokenizers as the backbone for each modality input. This approach is logical given the significantly different nature of the input data. For example, in speech processing, researchers typically work with a very high-resolution input (16 kHz waveform), which comprises hundreds of thousands of samples for typical utterances. To handle this, a multilayer convolutional neural network is necessary to transform it. This is not the case in NLP where word input sequences have much lower resolution and can be directly embedded in distributional space using a lookup table. The difference in type of data also influences how the authors of Data2Vec masked the input in order to create a sufficiently challenging learning task for the models. In NLP, removing individual words poses

a sufficiently challenging task as word tokens are often highly discrete. For speech, it is necessary to mask spans since adjacent audio samples are highly correlated, and masked at too short of a sample would allow the model to cheat by replicating local patterns. The same is for vision, patches too small would allow for models to simple copy nearby pixels while patches too large can corrupt too much valuable visual context, an appropriate patch size of pixel is fundamental to the success of representation learning.

In addition, in the base Data2Vec we use, researchers only trained the models with datasets within their own modalities prior to benchmarking. Although Data2Vec in its current state unifies the learning algorithm yet still learns representations individually from separate corpus for each modality, the researchers believe designing a single algorithm which facilitates the possibility to make future multi-modal learning simpler and more effective could lead to models that understand the world better through multiple modalities. Benchmarking extensively on three separate modalities with the base models, researchers observe Data2Vec outperform a variety of both supervised and self-supervised models (Baevski et al. 2022).

While the student-teacher part of the framework is built on a basic Transformer structure (Vaswani et al. 2023), the exact encoding process for each modality take inspiration from a variety of sources. A brief overview of the different preprocessing, masking, and pretraining pipeline are as follow:

3.1.1 Data2Vec-text. Data2Vec-text is heavily inspired by unimodal SSL text models mentioned in 2.2.1 like BERT(Devlin et al. 2019). The specific backbone model of Data2Vec Text is built upon RoBERTa's (Liu et al. 2019) implementation on fairseq (Ott et al. 2019), a Pytorch-based sequence modelling toolkit. Training data is first tokenized using a byte-pair encoding (Sennrich, Haddow, and Birch 2016) of 50K types. In keeping with the prediction of masked information paradigm across modalities, Data2Vec text uses the masking strategy of BERT but not the next-sentence prediction task (Devlin et al. 2019). 15% of tokens are selected, out of which 80% are replaced by a learned mask token, 10% are left unchanged and 10% are replaced by randomly selected vocabulary token. The text backbone model is pretrained on the same data as BERT, BooksCorpus (Zhu et al. 2015) and English Wikipedia, then fine-tuned on a similar array of downstream tasks (Devlin et al. 2019).

3.1.2 Data2Vec Vision. Data2Vec Vision takes after an array of unimodal SSL vision models including: Contrastive Learning models (Chen et al. 2020), Self-Distillation models (Caron et al. 2021; Oquab et al. 2024), and Masked Token Prediction models (Bao et al. 2022). Data2Vec starts by creating augmented images of randomly applied resized image crops, horizontal flipping, and color jittering commonly used in DML vision models, for example SimCLR (Chen et al. 2020). By first creating augmented versions of the original input images then masking out 16x16 pixels, images of 224x224 pixels as patches of 16x16 pixels. Each patch is linearly transformed and a sequence of 196 representations is input to a standard Transformer. This output then follows (Bao et al. 2022) by masking blocks of multiple adjacent patches where each block contains at least 16 patches with a random aspect ratio, a 60% of total patches are randomly selected to have the features set to a constant. The vision backbone models are first pretrained on ImageNet-1K (Deng et al. 2009) without being given any labels, then fine-tuned on the same dataset through image classification task using the labeled version of the same benchmark.

3.1.3 Data2Vec Audio. Data2Vec Audio is inspired by SSL works in audio processing that utilize either Contrastive Methods (Schneider et al. 2019) or Masked Token Prediction methods (Baevski et al. 2020). Using Wav2Vec2's (Baevski et al. 2020) audio preprocessing technique, the authors of Data2Vec created the *AutoProcessor.from_pretrained("facebook/data2vec-audio-base-960h")* class implemented through fairseq (Ott et al. 2019). It contains seven temporal convolutions with 512 channels, with strides of (5,2,2,2,2,2,2) and kernel widths of (10,3,3,3,3,2,2). Between each sample, this results in an encoder output frequency of 50 Hz with a stride of about 20ms, and a receptive field of 400 input samples or 25ms of audio. The masking strategy for the Base model also mirrors that of (Baevski et al. 2020): randomly select 6.5% of all time-steps as starting points and then mask the following ten time-steps. This approach ends up masking around 49% of all time-steps in a typical training sequence. Data2Vec is trained on Librispeech (Panayotov et al. 2015), a 960-hour speech audio dataset sampled at 16khz consisting of read English audiobooks. To account for the difference in sampling rate, Librispeech is encoded using a multi-layer 1-D convolutional neural network that maps 16 kHz waveform to 50 Hz representations, a function retained by the custom AutoProcessor class.

3.2 MUStARD++ Dataset

Taking inspiration from the explicit and implicit emotion labelling schema from (Chauhan et al. 2020), authors of "A Multimodal Corpus for Emotion Recognition in Sarcasm" make publicly available an updated and improved dataset MUStARD++ (Ray et al. 2022), which builds upon the corpus of MUStARD (Castro et al. 2019). Addressing the limited size and imbalanced sarcastic class of the original MUStARD, MUStARD++ contains 1202 samples in total, including the 690 videos from the original MUStARD, with newly selected videos from Big Bang Theory (seasons 9-13) and Silicon Valley. 50% of the video are sarcastic and 50% are not, with each sample similarly made up of the text modality (transcriptions to the spoken word), the audio and vision modality (a video clip). A rigorous process of interannotator agreement is conducted to provide new labels and contest some previously mislabelled instances in MUStARD. In addition to the explicit and implicit emotion labels (Chauhan et al. 2020), all samples are also annotated with type of sarcasm and valence/arousal scores.

As the primary task of MUStARD++ is to identify the explicit and implicit emotions in sarcastic utterances[1], researchers design a multiclass emotion classification model in which modalities are fused together in pairs using the attention mechanism, based on the collaboration gating architecture inspired by(Liu et al. 2020). However, the authors confirm that the same classifier framework also works for emotion regression task (predicting valence/arousal scores) and binary sarcasm detection. For the task of binary sarcasm detection, authors of MUStARD++ do not provide the models with the newly added emotion labels during training in contrast to (Chauhan et al. 2020). A detailed description of the model architecture is in section 3.6.

1 For the emotion recognition task in sarcastic utterances, the models are first trained on MELD (section 2.1.1) and IEMOCAP(Busso et al. 2008), a 12-hour corpus of audiovisual data, including video, speech, motion capture of face, text transcriptions (Poria et al. 2019; Busso et al. 2008). Videos from MELD and IEMOCAP are labelled with explicit emotion, implicit emotions, valence, activation, and dominance. However, due to IEMOCAP containing little to no sarcastic utterances, models even after pretraining only achieved F1-score of 40.8% on explicit emotion classification and 33.3% on implicit emotion classification.

For the text modality, researchers of MUStARD++ used BART-large for feature extraction (Lewis et al. 2019). The authors choose the average of the last four representations of the BART model to use as the text input, with a shape of d_t = 1024. For the audio modality, the researchers first sampled .wav files from the video with the audio signal at 22.5KHz. After performing noise and laughter extraction with Librosa (McFee et al. 2015), to restrict final feature tensors to a constant size despite their varying length in audio input, each audio is segmented and each segment of audio is layered on top of one another to achieve a consistent length. They then extract low-level audio features in the form of MFCC and Mel-Spectogram from Librosa (McFee et al. 2015), and prosodic features from OpenSMILE (Eyben, Wöllmer, and Schuller 2010). The shape of tensors from these three audio features are d_m = 128 , d_s = 128 , and d_p = 35, with the final input audio feature vector being a direct concatenation of the three, d_a = 291. For the vision modality, researchers first use the popular keyframe extraction open-source library Katna [2] to first select keyframes from either utterance or context, pass the keyframes into the a Resnet-152, and store the pool5 layer representations as the visual input. The size of final video feature representation is d_v = 2048.

Similar to previous studies based on (Castro et al. 2019), researchers perform exhaustive ablation studies by withholding certain modalities, speaker information, or the context to an utterance (mentioned in more detail in 3.6.3). Predicting with extracted features of three modalities without pretraining, the proposed binary sarcasm classification model achieves a new SOTA results of 74.2% F1-score on the original MUStARD dataset, and the highest F1-score of 70.3% in certain ablation settings on the extended MUStARD++ dataset.

3.3 Data Cleaning

During the exploratory data inspection of the MUStARD++ dataset, several issues within the original files were identified. Specifically, a number of video files have repeated or corrupted filenames that do not correspond to the 'KEY' value given to each utterance. By cross-referencing these videos to the values within the 'text' column and 'duration' column of the spreadsheet, all videos are able to find an appropriate match to an entry. Additionally, several rows within the dataset that contained incorrectly entered values are also identified in this process, whose value we also update accordingly. For example, row 2596 in the 'KEY' column where a unique tag that corresponds to a video clip is supposed to be, a string of 'DISGUST' (one of the implicit emotion labels) is stored instead. After identifying the video that corresponds with this row, the 'KEY' value is set to be '1_S11E03_067_u'. In the end, we end up with 1202 videos of utterances, 1202 videos of context, and 6041 rows in the spreadsheet, ensuring data integrity and reliability for subsequent analysis and processing.

3.4 Preprocessing and Tokenization

As the original MUStARD++ paper use three expert models in their own modality that produced tensors of vastly different size and nature, we devise a number of necessary changes at the preprocessing and tokenization stage (section 3.4.1, 3.4.2, 3.4.3) to appropriately adapt to the properties of Data2Vec, so as to maximize the potential efficacy of learning cross-modal features. We also explore a few variations in preprocessing and

2 https://katna.readthedocs.io/en/latest/

tokenization, whose effectiveness are briefly examined during pretesting (section 3.6.2, 4.1).

3.4.1 Text Modality. Tokenization of text is straightforward for Data2Vec, but we decide to experiment with 2 different ways to pass the transcription to the tokenizer. As Data2Vec-text utilizes the same tokenization scheme as RoBERTa, we decide to explore the effectiveness of using special tokens for better information representation in the case of multimodal dialogue sarcasm settings. The first input technique concatenates spoken sentences by alternating speakers within the same context scene with the RoBERTA special separator token </s>, which joins every alternation between before being passed to the Roberta tokenizer. This may help indicate the exchange nature of the sarcasm in dialogue form. As for the single utterance sentence, it is passed directly to the tokenizer. The second input technique adopts basic concatenation technique without adding in the separator token. Pretesting experiment details of these two input techniques can be found in section 3.6.2 and findings in section 4.1. All text input is processed by AutoTokenizer.from_pretrained("facebook/data2vec-text-base").

For text modality, we predict that the unimodal Data2Vec model may perform slightly worse than MUStARD++ text modal. This is due to the difference in Data2Vec-text embeddings compared to BART-large (Lewis et al. 2019) embeddings. On the GLUE benchmark, BART-large generally outperforms data2vec by a small margin, which may contribute to its superior performance in feature extraction. In addition to being a SSL model tuned via both BERT-like bidirectional encoding tasks as well as GPT-like autoregressive decoding tasks, BART-large's (Lewis et al. 2019) training corpus could potentially be more diverse [3] than that of Data2Vec-text, which is only pretrained on a small subset of the RoBERTa training corpus (Books Corpus and English Wikipedia) (Baevski et al. 2022).

3.4.2 Audio Modality. For audio modality, audio is extracted from the videos and stored in .wav format, with a sampling rate set to 16kHZ. Instead of the 22.5 khz used in MUStARD++, we decide to use the sampling rate the Data2Vec audio model (Baevski et al. 2022) is trained on as it would more closely reflect the granularity in which Data2Vec has learned its representations. Inspired by MUStARD++'s audio segmentation approach, two separate approaches to further preprocess audio are investigated.

For approach 1, we replicate the segment and layering technique of MUStARD++. As there is no specific detail as to the exact audio segmentation interval in the original paper, a length of 10 second intervals for the audio is decided by referring to the average and max length of both utterance and context videos. These statistics are included in Table 1. All audios shorter than ten second are padded to 10 seconds, and all audios longer than ten second have their segments layered on top of one another. This ensures the maximum amount of information is retained while making all inputs the same length.

For approach 2, we do not conduct segmentation and layering of the audio input and directly pass audios of varying length down the pipeline. This results in audio files between the range of 1 second and 60 seconds; while this avoids the issue of overlapping audio from the first approach, it relies heavily on a newly implemented attention mechanism (section 3.6) and the feature extractor to be able to scale effectively

3 BART (Lewis et al. 2019) does not specify its pretraining corpus in its paper.

	Utterance	Context
Unique Words	2695	4891
Avg. length (tokens)	11.3	7.8
Max length (tokens)	65	47
Max duration (seconds)	20	56
Avg. duration (seconds)	4.19	8.69

Table 1
Statistics of Sentences in the MUStARD++ Dataset

when processing signals of very different lengths. Pretesting experiment details of these two preprocessing techniques are described in section 3.6.2 and findings in section 4.1.

One notable difference is that authors of MUStARD++ perform noise and laughter removal with Librosa before feature extraction. As the vocal plot separation technique is officially deprecated by Librosa (McFee et al. 2015) as of version 0.10.1 citing non-competitiveness, we do not perform noise and laughter remover. All audio input is processed by `AutoProcessor.from_pretrained("facebook/data2vec-audio-base-960h")`.

For audio modality, we hypothesize while MUStARD++'s approach of low level audio features may capture vocal details that are not limited to spoken content, the demonstrated word-error-rate quality of Data2Vec-audio may allow it to identify the spoken content better than low level features.

3.4.3 Video Modality. In our experiments, a combination of preprocessing methods is used. To reduce computational load and maintain a degree of comparability to the results of MUStARD++, we also select keyframes with Katna [4] before feature extraction. Katna selects frames that are sufficiently different from previous ones and determines it as a *keyframe*, the representative frames of a video stream, or the frames that provide the most accurate and compact summary of the video content. This is achieved by an ensemble of image processing techniques: absolute differences in LUV (l:lightness, uv: chromaticity coordinates) colorspace, brightness score filtering of extracted frames, entropy/contrast score filtering of extracted frames, K-Means Clustering of frames with image histogram, and selection of best frame from clusters based on and variance of laplacian (image blur detection). A problem arises when some videos (42 utterance videos and 3 context videos) could not be processed by Katna due to their extremely short duration. For these videos, an older preprocessing strategy as outlined in the MUStARD (Castro et al. 2019) paper (extracting every single frame and averaging out its final feature representations across the time dimension) is used instead. Since these clips are relatively short and contain very few frames—-most of which have similar visual information—-the RGB values of all frames are averaged over the video's duration. This results in an output image that represents the combined information from all the frames stacked on top of each other. After the image for each utterance and context (1202 images for utterance and 1202 images for context) has been acquired, all images are processed by AutoTokenizer.from_pretrained("facebook/data2vec-vision-base").

4 https://katna.readthedocs.io/en/latest/

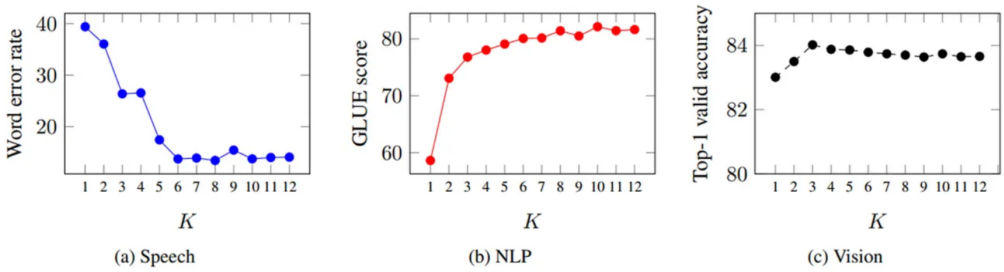


Figure 1
Top-K Layers Teacher Network Prediction Targets Impact on Final Representations Performance on Downstream Tasks, figure from (Baevski et al. 2022)

For vision modality, we predict that as on ImageNet-1K (Deng et al. 2009) Data2Vec vision reach a top-1 accuracy of 86.6% and Resnet-152 (He et al. 2015) a top-1 accuracy of 78.5%, unimodal Data2Vec models may perform better than the vision-only MUStARD++ models.

3.5 Feature Extraction

For each modality, the tokenized information is passed onto its corresponding model. Researchers of Data2Vec observe (Baevski et al. 2022) selecting different top K blocks of the teacher network as prediction targets has a sizeable impact on final benchmarking results for downstream tasks, as shown in Figure 1. Based on these metrics, for the base model of Data2Vec, researchers ultimately select K = 10 for text, K = 8 for speech, and K = 6 for vision. This proves that depending on the K value selected for each model, a significantly different mix of local and non-local meaningful representations could be situated across multiple layers. Inspired by using the mean of last four layers of BART in MUStARD++ for text (Ray et al. 2022) to capture more rich representations in deeper layers, we decide to experiment with applying the same technique to the audio and visual modality with Data2Vec. While top K layer as prediction target during training has no direct impact on which layers we select to use as final representations during feature extraction (as we use the Huggingface implementation which contains the student model), we believe the results to still be of experimental value. After extensive experimentation during pretesting, we decide to conduct final sarcasm detection ablation studies on two sets of models, one tuned with final layer representations for all modalities and one tuned with mean of last four layer representations for all modalities. For the mean of last four layers version of the final data, all feature tensors are in the shape [1, 768]. For the final layer version of the data, feature vectors output are respectively in the shape of [1, 768] for text, [1,197,768] for image, and ranging between [1, 18, 768] and [1, 2813, 768] for audio.

3.6 Model Architecture and Experiments

3.6.1 Binary Sarcasm Classification Model. We decide to use the same collaborative gating architecture as MUStARD++ [5], which was inspired by (Liu et al. 2020). We describe the structure of the classification model belonging to the early fusion type, which consists of a **fusion module** and a **prediction module**, in brief below.

In the **fusion module**, each input (features of either utterance or context, in one of the three modalities) is transformed to a shared embedding space via linear layers followed by ReLU activation and batch normalization. An attention mechanism then computes attention scores between the utterance and context embeddings, as well as between different modalities. The shared embeddings for utterances are updated using the attention scores computed with respect to their own contexts and other modalities. Finally, the updated shared embeddings from all modalities are concatenated along with speaker embeddings (speaker information passed in as a one-hot vector). The fully-fused embeddings then are then passed onto the **prediction module**, which is made up of a series of fully connected layers with batch normalization, ReLU activation, and dropout, before the final layer with two neurons in the case of binary sarcasm classification.

One addition to the training process is a basic **sequence-attention mechanism** we implemented. For all previously mentioned tensors with three dimensionalities (section 3.5), the *apply_attention* function (Listing 1) is used to reduce the extra dimensionality before passing into the classification model.

The function takes a single argument tensor which has the shape [*batch_size*, *seq_len*, *hidden_dim*], where *batch_size* is the number of samples in a batch, *seq_len* is the length of the sequence, and *hidden_dim* is the dimension of the hidden states. A linear transformation is applied to each hidden state vector in the sequence. This transformation reduces the dimensionality of each vector to a single value, producing a score for each position in the sequence. The attention scores are then normalized using the softmax function along the sequence dimension, producing a probability distribution over the sequence positions, giving the attention weights tensor the same shape [*batch_size*, *seq_len*, *1*]. Lastly, the output tensor is produced is by a weighted sum calculated across the sequence dimension, resulting in a tensor of shape [*batch_size*, *hidden_dim*].

Listing 1
Sequence-Attention Mechanism

```
def apply_attention(tensor):

    attention_layer = torch.nn.Linear(768, 1)
    attention_scores = attention_layer(tensor)
    attention_weights = F.softmax(attention_scores, dim=1)
    attended_tensor = (tensor * attention_weights).sum(dim=-2)
    return attended_tensor
```

3.6.2 Pretesting and Hyperparameter Search. To reduce the overall computational load while still exploring how input, preprocessing, or feature extraction methods mentioned

5 MUStARD++'s binary sarcasm classification models are not pretrained, unlike its emotion classification and emotion regression models, which are pretrained on the IEMOCAP dataset (section 3.2). As emotion recognition in sarcastic setting is not relevant in our studies, we will also not conduct any form of pretraining for our binary sarcasm classification experiments.

previously could impact SSL transformers' performance as feature extractors, we conduct a two-stage pretesting with hyperparameter search on parts of the full dataset. Ablation studies configurations during pretesting are identical to those described in the final sarcasm detection experiments (section 3.6.3).

During stage 1, a toy dataset of 200 utterances of stratified sarcastic/non-sarcastic videos is randomly sampled and the following variations are tested:

1. using direct concatenation for context sentences or concatenating them with the special Roberta separator token <s/> (as mentioned in section 3.4.1)
2. using layered audios of consistent length vs. using unlayered audios of varying lengths (as mentioned in section 3.4.2)
3. using final layer representations from the Data2Vec-audio-960h and Data2Vec-vision-base model vs. using the mean of the last four layers for the audio and video modality (as mentioned in section 3.5)

During stage 2, we build upon the findings of stage 1 (section 4.1) and sample 450 videos with stratified sarcastic/non-sarcastic randomly to re-examine the final layer vs. mean of last four layer approach, reporting these results also in section 4.1.

For all pretesting, we replicate the hyperparameter search range of the original MUStARD++, examining dropout values of [0.2, 0.3, 0.4], learning rates of [0.001, 0.0001], batch sizes of [32, 64, 128], shared embedding dimensions of [2048, 1024], and projection embedding dimensions of [1024, 256]. In the end based on all findings from the pretesting experiments, we decide to create two versions of the full dataset, one based on final layer representations and the other based on mean of last four layer representations.

3.6.3 Final Sarcasm Detection. With the two versions of feature tensors produced and best hyperparameters for each ablation configuration selected (section 3.5, 4.1), we perform a five fold cross-validation binary sarcasm prediction results across all ablation configurations, identical to those in MUStARD++ (Ray et al. 2022). These include 7 kinds of modality combination (V ,T , A, VT, TA, VA, VTA) [6], 2 kinds of speaker information setup (with or without), and 2 kinds of context setup (with or without). In conjunction with the two sets of feature tensors, a total of 56 models are trained, and we report these results in section 4.2.

4. Results and Discussions

4.1 Pretesting/Hyperparameter Search

From stage 1 pretesting on the toy dataset, we observe that:

1. Direct concatenations perform mostly par with separator-populated concatenations. We decide to conduct the final evaluation based on the separator token approach, as we hope the turn based nature of dialogue to increases in importance as the dataset gets larger.

6 V = Vision, T = Text, A = Audio

2. Prelayered audios perform mostly on par with non-layered audio but with few occasions of slightly lower F1-score. We decide to conduct our final experiments with the non-layered audios for two reasons: 1) it ensures the least amount of human input prior to training thus preventing the segmentation length from becoming a future variable when evaluating the effectiveness of Data2Vec, and 2) it capitalizes on SSL's potential to capture more meaningful representations with long-range dependencies when it is not confined to a set duration.
3. The final layer representations perform slightly better on average than the mean of the last four layer representation except in a few cases, but overall results remain inconclusive.

During stage 2 pretesting, as a further exploration of variation 3 in the previous pretesting, we once again perform hyperparameter search on a bigger 450 video sample. We observe that in more than half of the ablation settings, final layer representations perform a considerable amount better (Appendix A). However, our ultimate decision to extract both versions of feature from the full dataset is motivated by two reason:

1. While preliminary exploration does not favor last four layer representations over final layer representations, compared to the other two variations tested in stage 1, the comparison of final vs. four has more implication in terms of examining the nature of Data2Vec's embeddings in regards to capturing non-local, potentially cross-modal features. As such, we want to observe if this may be an emergent property when applied to a sufficiently large training set.
2. While averaging the last four layer technically could cause positional issues especially in audio or vision modality, this method has a benefit of maintaining feature tensors to a consistent size of [1, 768], regardless of modality or length of an utterance. This means a significantly reduced computational load at training time, as we wouldn't have to apply the sequence-attention mechanism on the fly as the classification model's weights and biases update. We further discuss this in section 5.1 after evaluating final results.

The best hyperparameters are selected at this stage and used for final sarcasm detection. (Appendix A)

4.2 Final Sarcasm Detection Results

Table 3 and Table 4 contain all precision, recall, and F1 scores for models trained and tuned using the mean of last four layer representations, and Table 5 and Table 6 contain those for models trained and tuned using the representations extracted from the final layer of the models. These results contain the experiments of every single modality and context/speaker info combination done in our ablation studies, averaged across a five-fold cross validation. For reference purposes, we include ablations studies results to (Ray et al. 2022) in Table 2.

4.2.1 Unimodal Performances. When examining purely unimodal models, some models seem to struggle to capture important features that would help predict sarcasm more

Table 2
Ablation Studies Results from MUStARD++ (Ray et al. 2022)

	Speaker Independent						**Speaker Dependent**					
	w/o Context			**w Context**			**w/o Context**			**w Context**		
	P	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**	**P**	**R**	**F1**
T	67.9	67.7	67.7	69.3	69.2	69.2	69.4	69.3	69.3	70.2	70	70
A	63.9	63.5	63.6	64.3	64.1	64.1	65.3	65.2	65.2	65.0	64.9	64.9
V	59.5	59.4	59.4	60.3	60.0	60.0	61.8	61.7	61.7	61.6	61.4	61.5
T+A	68.8	68.6	68.7	**70.2**	**70.2**	**70.2**	**69.8**	**69.5**	**69.5**	69.2	69.1	69.1
A+V	65.7	65.4	65.5	67.5	67.3	67.4	64.9	64.5	64.5	64.2	64.0	64.0
V+T	68.2	68.1	68.1	67.9	67.6	67.6	69.1	69.0	69.0	69.4	69.1	69.1
T+A+V	**69.5**	**69.4**	**69.4**	69.6	69.5	69.6	69.6	69.3	69.3	**70.6**	**70.3**	**70.3**

	Speaker Dependent			Speaker Independent		
	Precision	Recall	F1-score	Precision	Recall	F1-score
T	59.0	59.0	59.0	57.1	57.1	57.0
A	61.7	61.6	61.6	57.2	57.2	57.1
V	62.3	62.3	62.2	61.3	61.3	61.2
TA	60.9	60.9	60.9	60.6	60.6	60.6
TV	59.2	59.2	59.2	59.4	59.4	59.3
AV	**62.4**	**62.4**	**62.4**	**63.4**	**63.4**	**63.3**
TVA	60.5	60.5	60.5	59.6	59.6	59.6

Table 3
Precision, Recall, F1 Scores of Models, Last Four Layer of Utterance

	Speaker Dependent			Speaker Independent		
	Precision	Recall	F1-score	Precision	Recall	F1-score
T	62.1	62.0	62.0	57.9	57.8	57.8
A	61.1	61.1	61.1	58.3	58.3	58.2
V	61.8	61.6	61.4	61.1	61.1	61.0
TA	62.7	62.7	62.6	56.7	56.7	56.6
TV	60.5	60.5	60.5	59.8	59.8	59.8
AV	61.8	61.8	61.8	**67.2**	**67.2**	**67.1**
TVA	**63.0**	**62.9**	**63.0**	62.6	62.6	62.6

Table 4
Precision, Recall, F1 Scores of Models, Last Four Layer of Utterance + Context

than others. Vision models seem to perform better than either the text or audio modality, which exists in stark contrast to MUStARD++ (Ray et al. 2022). For text models, we witness a fall in F1-score up to 10%, which is consistent with our hypothesis in 3.4.1 that Data2Vec-text embeddings may perform worse than BART-large (Lewis et al. 2019) embeddings due to previous performance on benchmarks, pretext task diversity, and training corpus size. For vision models, we witness comparative and even improved results compared to MUStARD++, which also corroborates our hypothesis in 3.4.3,

	Speaker Dependent			Speaker Independent		
	Precision	Recall	F1-score	Precision	Recall	F1-score
T	60.3	60.3	60.2	57.2	57.2	57.1
A	59.3	59.3	59.2	57.6	57.6	57.6
V	61.7	61.6	61.6	61.7	61.7	61.7
TA	61.5	61.5	61.3	56.7	56.6	56.6
TV	60.4	60.4	60.4	60.9	60.9	60.9
AV	**62.4**	**62.3**	**62.3**	**64.6**	**64.5**	**64.5**
TVA	60.2	60.1	60.1	63.8	63.8	63.8

Table 5
Precision, Recall, F1 Scores of Models, Final Layer of Utterance

	Speaker Dependent			Speaker Independent		
	Precision	Recall	F1-score	Precision	Recall	F1-score
T	60.0	60.0	60.0	57.6	57.6	57.5
A	61.5	61.5	61.5	58.4	58.4	58.4
V	60.8	60.8	60.8	59.1	59.1	59.1
TA	60.8	60.8	60.8	57.1	57.1	57.0
TV	62.5	62.5	62.4	58.2	58.1	58.1
AV	62.2	62.2	62.2	**65.6**	**65.6**	**65.6**
TVA	**64.5**	**64.5**	**64.5**	62.9	62.8	62.8

Table 6
Precision, Recall, F1 Scores of Models, Final Layer of Utterance + Context

that the better performance of Data2Vec-vision on ImageNet-1K (Deng et al. 2009) over Resnet-152 (He et al. 2015) is transferred to detecting visual cues of sarcasm. For audio, the drop in quality may be due to the training corpus of Data2Vec-audio. As Data2Vec is trained on Librispeech (Panayotov et al. 2015) which contains mostly read passages, the benefit it offers by being able to identify spoken content better ultimately does not outweigh the importance of non-spoken vocal cues that are better captured by low level audio features. [7] We represent the general trend of isolated performance by individual modality here as ($V_1 > A_2 > T_3$).

4.2.2 Trends in Modality Mixing. We observe that in bimodal models of Vision and Audio end up performing the best out of all modality combinations, surpassing that of trimodal models in the majority of the cases. This is mostly in line with how unimodal input perform, as combining the two most effective input theoretically would yield the best results. This is also the case in MUStARD++ (Ray et al. 2022) for text and audio.

7 According to MUStARD++, (Ray et al. 2022) researchers briefly experimented with the application of PASE, a problem-agnostic self-supervised audio transformer (Ravanelli et al. 2020), before ultimately using low level features from Librosa. Citing results to be suboptimal, PASE is dropped early on without being further explored upon. While authors similarly attributed to the limited size of MUStARD++ contributing to the difficulty for SSL methods to learn sufficient representations, they did not provide exact figures detailing PASE's performance in feature extraction. As such, we could not directly compare its effectiveness to Data2Vec-audio's.

However, one difference observed between MUStARD++ and our models is that the *dominance* of a modality when used as the single input doesn't translate directly to its performance in dual modality configurations. In the experiments of MUStARD++, high-impact modalities (descending order text, audio, vision) tend to fully dictate any models that use them in the fusion process. That is, when choosing a second modality to complement a fixed modality, the ranking of second ability based on T_1>A_2>V_3 always predicts the outcome of that model T_1A_2>T_1V_3>A_2V_3 (Table 2).

In comparison, while isolated modality performances in our ablation studies are V_1 > A_2 > T_3, dual models using text as one of the two modality sometimes perform better when the other input is audio, not vision (T_3A_2>T_3V_1). Despite unimodal vision predictions outperforming unimodal audio predictions consistently, only in about half cases does its collaboration with text (TV) prove superior to text with audio's (TA). This could allude to the fact that the text modality of transcriptions might have a quasi-semantic relation to the audio modality of human speech, as tokenization and training data of text transformers and audio transformers are more closely aligned, in contrast to vision transformers. While this does not suggest that actual cross-modal features are acquired and subsequently influenced the results, it does suggest that even with unified learning objectives, the exact selection of tokenization and training corpus could have implications further down the line when learned representations are utilized for more complicated downstream tasks.

4.2.3 Final Layer vs. Mean of Last Four Layer. We observe that the representations produced by the final layer and the mean of last four layers seem to perform mostly on par, with minor edge over each other with no observable pattern. This is in contrast to pretesting results (section 4.1) where final layer representations displayed a slight advantage. However, the hypothesized trend we hoped to see where as the training size of the model increases, the mean of last four layer representations may prove to in usefulness by capturing more non-local and high-dimensional abstractions (section 4.1) does not show up either. In short, while there is no explicit benefit in prediction to choose one over another, we believe that in future applications, the mean of last four layer representations produced by various Data2Vec models can be considered equally useful for downstream tasks as the final layer representations, as the former provide similar features learned than the latter but with much less computational load.

4.2.4 Additional Observations. While not the main focus of our study, we observe when given speaker information to help the model identify whether sarcasm is presence in an utterance, models generally improve in performance, which is consistent with previous research (Castro et al. 2019; Chauhan et al. 2020; Ray et al. 2022). However, when given context, models do not display a clear and consistent improve in performance like that of MUStARD++ (table 2). While some models show a slight increase in performance compared to their utterance-only counterparts, these improvements do not display an observable pattern even when taken into consideration different modality combinations.

5. Conclusion

5.1 Findings

In conclusion, regardless of either using mean of last four representations or final layer representations, we are not able to see Data2Vec surpass the performance of binary

sarcasm classification models in MUStARD++ that are built on the fusion of three expert feature extraction methods. Despite this, there does seem to be occasions where bimodal or trimodal models' performance change could be related to how the input modalities are processed by Data2Vec. This differs from MUStARD++ models, where the trend of text as a modality dominating other modalities carries over to dual and triple setups. While this is not definitive proof of Data2Vec's capability to learn meaningful features in one modality by being trained on data of a different modality, it might suggest additional training of Data2Vec on multimodal datasets may lead it to capture specific cross-modal cues in well-defined tasks.

Through our study, we observe that in the task of multimodal sarcasm detection, using modality-agnostic self-supervised transformers exclusively as feature extractors for all modalities does not improve the performance on the MUStARD++ dataset. Additionally, despite designed to be modality-agnostic in its shared learning objective, Data2Vec when applied to multimodal NLP task that engages all the modalities it has previously excelled on other benchmarks, there is no noticeable benefit as more modalities are combined. While there are minor improvements in specific bimodal or trimodal configurations over using MUStARD++'s fusion of three expert models, these are not very consistent and could be due to either the difference in preprocessing and feature extraction, but not unambiguously attributed to the capturing of modality-agnostic latent representations.

However, one key practical discovery is that in the case of using Data2Vec's pretrained knowledge to extract features, last four layer representations could be equally as useful as last layer representations. There provides an alternative for future researchers to reduce overhead when leveraging learned features from large, multimodal pretrained models by selecting specific layers that may contain equal amounts of useful latent representations at a much lower computational cost.

5.2 Discussions

As the three backbone of Data2Vec are pretrained by separate modalities, their exact extracted features are still largely restrained unimodally to what they learned from their pretraining corpus. While through our experiments we cannot prove the definitive interaction between Data2Vec's embeddings across modalities, we do observe some potential in certain cases (e.g. text and audio). However, due to limited scope of the project and computational constraint, we could not experiment with a wider variety or combination of SSL feature extractors based on different learning mechanism and modalities to fully investigate this effect.

This research is thus considered a humble addition to the scarce literature of using SSL on all modalities for feature extraction in the MUStARD (Castro et al. 2019) dataset family. In (Bhosale et al. 2023), researchers experiment with finetuning Wav2Vec2 (Baevski et al. 2020) on a speech emotion recognition task (IEMOCAP, (Busso et al. 2008)) before extracting features from MUStARD++, and achieve a 67.5% on audio-only, which proves that finetuning SSL on tasks relevant to sarcasm detection does yield improved performance. This suggests in future research, finetuning Data2Vec-text, Data2Vec-vision, and Data2Vec-audio on modality-specific tasks that are relevant to sarcastic language usage would also yield better results, and potentially paint a fuller picture the interaction between Data2Vec's embeddings across modalities.

Table 1
Best hyperparameters (dropout, learning rate, batch size, shared embedding dim., projection embedding dim.) per ablation setting

	Mean of Last Four Layer Representations	Final Layer Representations
SCVTA	(0.2, 0.001, 128, 1024, 256) with loss: 0.599856 and F1 score of 0.695293	(0.2, 0.001, 32, 2048, 1024) with loss: 0.593293 and F1 score of 0.691069
CVTA	(0.4, 0.001, 32, 2048, 256) with loss: 0.626675 and F1 score of 0.663877	(0.3, 0.001, 64, 1024, 1024) with loss: 0.620705 and F1 score of 0.656254
SVTA	(0.4, 0.001, 128, 1024, 1024) with loss: 0.636428 and F1 score of 0.650793	(0.2, 0.0001, 32, 1024, 1024) with loss: 0.600724 and F1 score of 0.718959
VTA	(0.4, 0.001, 64, 2048, 256) with loss: 0.626669 and F1 score of 0.679197	(0.3, 0.001, 32, 1024, 256) with loss: 0.635544 and F1 score of 0.656166
SCVT	(0.3, 0.001, 128, 1024, 256) with loss: 0.639577 and F1 score of 0.617164	(0.4, 0.001, 128, 2048, 256) with loss: 0.589899 and F1 score of 0.715344
SCVA	(0.2, 0.0001, 64, 1024, 256) with loss: 0.632554 and F1 score of 0.656166	(0.4, 0.001, 128, 2048, 256) with loss: 0.601528 and F1 score of 0.700492
SCTA	(0.3, 0.001, 128, 1024, 256) with loss: 0.630159 and F1 score of 0.656253	(0.4, 0.001, 128, 2048, 256) with loss: 0.593385 and F1 score of 0.714285
CVT	(0.4, 0.001, 64, 2048, 256) with loss: 0.628321 and F1 score of 0.654139	(0.3, 0.001, 128, 1024, 1024) with loss: 0.631456 and F1 score of 0.648244
CVA	(0.2, 0.001, 64, 1024, 256) with loss: 0.631378 and F1 score of 0.647382	(0.3, 0.001, 128, 2048, 256) with loss: 0.618955 and F1 score of 0.686811
CTA	(0.2, 0.001, 64, 1024, 1024) with loss: 0.623707 and F1 score of 0.701960	(0.3, 0.001, 64, 1024, 256) with loss: 0.633637 and F1 score of 0.655493
SVT	(0.4, 0.0001, 32, 2048, 1024) with loss: 0.635008 and F1 score of 0.679197	(0.2, 0.001, 64, 1024, 1024) with loss: 0.569263 and F1 score of 0.741793
SVA	(0.4, 0.001, 32, 2048, 1024) with loss: 0.627216 and F1 score of 0.655493	(0.4, 0.001, 64, 1024, 256) with loss: 0.637161 and F1 score of 0.678096
STA	(0.4, 0.001, 32, 2048, 256) with loss: 0.631316 and F1 score of 0.652087	(0.2, 0.0001, 64, 1024, 256) with loss: 0.629902 and F1 score of 0.670588
VT	(0.4, 0.0001, 128, 2048, 1024) with loss: 0.690382 and F1 score of 0.561536	(0.3, 0.001, 128, 2048, 256) with loss: 0.683520 and F1 score of 0.640404
VA	(0.3, 0.0001, 64, 2048, 256) with loss: 0.676442 and F1 score of 0.567567	(0.4, 0.001, 64, 2048, 1024) with loss: 0.624358 and F1 score of 0.663549
TA	(0.3, 0.0001, 64, 2048, 1024) with loss: 0.687365 and F1 score of 0.542857	(0.4, 0.001, 64, 1024, 256) with loss: 0.655634 and F1 score of 0.639215
SCV	(0.4, 0.001, 32, 2048, 256) with loss: 0.658720 and F1 score of 0.623529	(0.3, 0.0001, 64, 2048, 256) with loss: 0.605942 and F1 score of 0.679197
SCT	(0.2, 0.001, 64, 1024, 256) with loss: 0.652579 and F1 score of 0.639215	(0.4, 0.001, 128, 2048, 256) with loss: 0.592070 and F1 score of 0.718130
SCA	(0.2, 0.001, 128, 1024, 256) with loss: 0.619797 and F1 score of 0.671152	(0.3, 0.001, 64, 2048, 1024) with loss: 0.608287 and F1 score of 0.710070
CV	(0.4, 0.001, 64, 2048, 256) with loss: 0.639599 and F1 score of 0.663549	(0.4, 0.001, 64, 2048, 1024) with loss: 0.612227 and F1 score of 0.671794
CT	(0.2, 0.0001, 64, 1024, 256) with loss: 0.661873 and F1 score of 0.592855	(0.3, 0.0001, 64, 1024, 1024) with loss: 0.671378 and F1 score of 0.553351
CA	(0.2, 0.0001, 32, 2048, 1024) with loss: 0.643008 and F1 score of 0.6322513	(0.4, 0.001, 128, 1024, 256) with loss: 0.634590 and F1 score of 0.624633
SV	(0.4, 0.001, 32, 2048, 1024) with loss: 0.640004 and F1 score of 0.647382	(0.4, 0.001, 64, 2048, 1024) with loss: 0.619072 and F1 score of 0.674157
ST	(0.4, 0.001, 32, 2048, 1024) with loss: 0.640004 and F1 score of 0.647382	(0.2, 0.001, 64, 2048, 256) with loss: 0.615610 and F1 score of 0.707532
SA	(0.3, 0.001, 64, 1024, 256) with loss: 0.604243 and F1 score of 0.702471	(0.3, 0.0001, 128, 2048, 1024) with loss: 0.599733 and F1 score of 0.741442
V	(0.3, 0.0001, 32, 2048, 256) with loss: 0.696544 and F1 score of 0.569023	(0.4, 0.001, 32, 2048, 1024) with loss: 0.615692 and F1 score of 0.671152
T	(0.4, 0.0001, 128, 2048, 256) with loss: 0.693422 and F1 score of 0.601343	(0.4, 0.001, 64, 2048, 256) with loss: 0.662941 and F1 score of 0.625341
A	(0.3, 0.0001, 64, 1024, 1024) with loss: 0.648761 and F1 score of 0.626219	(0.3, 0.001, 64, 1024, 256) with loss: 0.665813 and F1 score of 0.599582

Another aisle of improvement to look into is to better utilize visual information. In order to reduce computational load, our research inherits the usage of keyframe extraction often used in multimodal sarcasm detection works, instead of processing the entire video (Ray et al. 2022). However, with the advent of robust video transformers like ViViT (Arnab et al. 2021), replacing a static image with the representations of the sequential data of a video could prove to be much more beneficial for future works as a visual input. As aforementioned in 3.2, Bhosale et al. (2023) already deploys ViFi-Clip (Rasheed et al. 2023) to better capitalize on the sequential visual information, but ViFi-Clip has a separate pretext task from Data2Vec. For future researchers, it would be particularly interesting to explore the cross interaction of using an SSL model that predict latent contextualized representations from sequential visual data, alongside Data2Vec (or other SSL with similar pretext learning objectives).

Lastly, one of the longstanding problem for multimodal sarcasm detection is that in certain types of sarcastic utterances, specific modalities may offer little to no prediction value not just for the model, but also not useful for interpreting sarcasm for human. However, when performing ablation studies there are no gold standard labels to which to compare whether the supplementation of a given modality could have "confused" the model. Extra annotation added to the MUStARD++ dataset that marks each isolated modality for a given sarcastic instance as "contributes to sarcasm" or "does not contribute to sarcasm" could help future researchers address the multimodal sarcasm detection problem in a more fine-grained manner.

Appendix

Appendix A: Hyperparameter Search Results

Hyperparameter search between dropout values of [0.2, 0.3, 0.4], learning rates of [0.001, 0.0001], batch sizes of [32, 64, 128], shared embedding dimensions of [2048, 1024], and projection embedding dimensions of [1024, 256] based on minimizing loss. (V = Vision, T = Text, A = Audio, S = Speaker Information Included, C = Context Included)

Appendix B: MUStARD++ Dataset Repository

The MUStARD++ dataset repository contains the .csv file used in this paper and a link to all videos stored on Google Drive. It also contains the original classification model and training script. `https://github.com/cfiltnlp/MUStARD_Plus_Plus/tree/main`

Appendix C: Project Repository

All code mentioned in this paper including: A) preprocessing script and feature extraction script, B) pretesting/hyperparameter search, C) modified classification model script, and D) final training and evaluation script is stored in a GitHub repository. Additionally, the 5 fold of data of both splits are also stored at this repository for reference.

`https://github.com/patwuch/mustard_sarcasm`

References

Abdullah, Sharmeen M Saleem Abdullah, Siddeeq Y Ameen Ameen, Mohammed AM Sadeeq, and Subhi Zeebaree. 2021. Multimodal emotion recognition using deep learning. *Journal of Applied Science and Technology Trends*, 2(01):73–79.

Arnab, Anurag, Mostafa Dehghani, Georg Heigold, Chen Sun, Mario Lučić, and Cordelia Schmid. 2021. Vivit: A video vision transformer.

Baevski, Alexei, Arun Babu, Wei-Ning Hsu, and Michael Auli. 2023. Efficient self-supervised learning with contextualized target representations for vision, speech and language.

Baevski, Alexei, Wei-Ning Hsu, Qiantong Xu, Arun Babu, Jiatao Gu, and Michael Auli. 2022. data2vec: A general framework for self-supervised learning in speech, vision and language.

Baevski, Alexei, Henry Zhou, Abdelrahman Mohamed, and Michael Auli. 2020. wav2vec 2.0: A framework for self-supervised learning of speech representations.

Balestriero, Randall, Mark Ibrahim, Vlad Sobal, Ari Morcos, Shashank Shekhar, Tom Goldstein, Florian Bordes, Adrien Bardes, Gregoire Mialon, Yuandong Tian, Avi Schwarzschild, Andrew Gordon Wilson, Jonas Geiping, Quentin Garrido, Pierre Fernandez, Amir Bar, Hamed Pirsiavash, Yann LeCun, and Micah Goldblum. 2023. A cookbook of self-supervised learning.

Bao, Hangbo, Li Dong, Songhao Piao, and Furu Wei. 2022. Beit: Bert pre-training of image transformers.

Bharti, Santosh Kumar, Rajeev Kumar Gupta, Prashant Kumar Shukla, Wesam Atef Hatamleh, Hussam Tarazi, and Stephen Jeswinde Nuagah. 2022. Multimodal sarcasm detection: a deep learning approach. *Wireless Communications and Mobile Computing*, 2022(1):1653696.

Bhosale, Swapnil, Abhra Chaudhuri, Alex Lee Robert Williams, Divyank Tiwari, Anjan Dutta, Xiatian Zhu, Pushpak Bhattacharyya, and Diptesh Kanojia. 2023. Sarcasm in sight and sound: Benchmarking and expansion to improve multimodal sarcasm detection.

Bouazizi, Mondher and Tomoaki Otsuki Ohtsuki. 2016. A pattern-based approach for sarcasm detection on twitter. *IEEE Access*, 4:5477–5488.

Brandes, Nadav, Dan Ofer, Yam Peleg, Nadav Rappoport, and Michal Linial. 2022. Proteinbert: a universal deep-learning model of protein sequence and function. *Bioinformatics*, 38(8):2102–2110.

Busso, Carlos, Murtaza Bulut, Chi-Chun Lee, Abe Kazemzadeh, Emily Mower, Samuel Kim, Jeannette N Chang, Sungbok Lee, and Shrikanth S Narayanan. 2008. Iemocap: Interactive emotional dyadic motion capture database. *Language resources and evaluation*, 42:335–359.

Camp, Elisabeth. 2012. Sarcasm, pretense, and the semantics/pragmatics distinction. *Noûs*, 46(4):587–634.

Caron, Mathilde, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. 2021. Emerging properties in self-supervised vision transformers.

Castro, Santiago, Devamanyu Hazarika, Verónica Pérez-Rosas, Roger Zimmermann, Rada Mihalcea, and Soujanya Poria. 2019. Towards multimodal sarcasm detection (an _obviously_ perfect paper). *arXiv preprint arXiv:1906.01815*.

Caucci, Gina M and Roger J Kreuz. 2012. Social and paralinguistic cues to sarcasm. *Humor*, 25(1):1–22.

Chauhan, Dushyant Singh, Dhanush S R, Asif Ekbal, and Pushpak Bhattacharyya. 2020. Sentiment and emotion help sarcasm? a multi-task learning framework for multi-modal sarcasm, sentiment and emotion analysis. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 4351–4360, Association for Computational Linguistics, Online.

Chauhan, Dushyant Singh, Gopendra Vikram Singh, Aseem Arora, Asif Ekbal, and Pushpak Bhattacharyya. 2022. An emoji-aware multitask framework for multimodal sarcasm detection. *Knowledge-Based Systems*, 257:109924.

Chen, Ting, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. 2020. A simple framework for contrastive learning of visual representations.

Cortes, Corinna and Vladimir Vapnik. 1995. Support-vector networks. *Machine learning*, 20(3):273–297.

Das, Ringki and Thoudam Doren Singh. 2023. Multimodal sentiment analysis: a survey of methods, trends, and challenges. *ACM Computing Surveys*, 55(13s):1–38.

Deng, Jia, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. 2009. Imagenet: A large-scale hierarchical image database. In *2009 IEEE conference on computer vision and pattern recognition*, pages 248–255, Ieee.

Devlin, Jacob, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding.

Dosovitskiy, Alexey, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. 2021. An image is worth 16x16 words: Transformers for image recognition at scale.

Erfina, Adhitia, Aidah Sri Tamanin, Listianawati, Falentino Sembiring, Sudin Saepudin, Melawati, and CSA Teddy Lesmana. 2020. New approach of sarcasm detection in indonesian marketplace product review. In *2020 6th International Conference on Computing Engineering and Design (ICCED)*, pages 1–4.

Eyben, Florian, Martin Wöllmer, and Björn Schuller. 2010. opensmile – the munich versatile and fast open-source audio feature extractor. In *MM'10 - Proceedings of the ACM Multimedia 2010 International Conference*, pages 1459–1462.

Filik, Ruth, Hartmut Leuthold, Katie Wallington, and Jemma Page. 2014. Testing theories of irony processing using eye-tracking and erps. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 40(3):811.

Frenda, Simona et al. 2018. The role of sarcasm in hate speech. a multilingual perspective. In *Proceedings of the doctoral symposium of the xxxivinternational conference of the spanish society for natural language processing (sepln 2018)*, pages 13–17, Lloret, E.; Saquete, E.; Martínez-Barco, P.; Moreno, I.

Gadzicki, Konrad, Razieh Khamsehashari, and Christoph Zetzsche. 2020. Early vs late fusion in multimodal convolutional neural networks. In *2020 IEEE 23rd international conference on information fusion (FUSION)*, pages 1–6, IEEE.

Goodfellow, Ian. 2016. *Deep Learning*. MIT Press.

Grill, Jean-Bastien, Florian Strub, Florent Altché, Corentin Tallec, Pierre H. Richemond, Elena Buchatskaya, Carl Doersch, Bernardo Avila Pires, Zhaohan Daniel Guo, Mohammad Gheshlaghi Azar, Bilal Piot, Koray Kavukcuoglu, Rémi Munos, and Michal Valko. 2020. Bootstrap your own latent: A new approach to self-supervised learning.

He, Kaiming, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2015. Deep residual learning for image recognition.

Helal, Nivin A, Ahmed Hassan, Nagwa L Badr, and Yasmine M Afify. 2024. A contextual-based approach for sarcasm detection. *Scientific Reports*, 14(1):15415.

Hsu, Wei-Ning, Benjamin Bolte, Yao-Hung Hubert Tsai, Kushal Lakhotia, Ruslan Salakhutdinov, and Abdelrahman Mohamed. 2021. Hubert: Self-supervised speech representation learning by masked prediction of hidden units.

Hussain, Jabbar. 2019. Deep learning black box problem.

Kannangara, Sandeepa. 2018. Mining twitter for fine-grained political opinion polarity classification, ideology detection and sarcasm detection. In *Proceedings of the Eleventh ACM International Conference on Web Search and Data Mining*, pages 751–752.

Lewis, Mike, Yinhan Liu, Naman Goyal, Marjan Ghazvininejad, Abdelrahman Mohamed, Omer Levy, Ves Stoyanov, and Luke Zettlemoyer. 2019. Bart: Denoising sequence-to-sequence pre-training for natural language generation, translation, and comprehension.

Li, Zuhe, Yangyu Fan, Bin Jiang, Tao Lei, and Weihua Liu. 2019. A survey on sentiment analysis and opinion mining for social multimedia. *Multimedia Tools and Applications*, 78:6939–6967.

Liu, Yang, Samuel Albanie, Arsha Nagrani, and Andrew Zisserman. 2020. Use what you have: Video retrieval using representations from collaborative experts.

Liu, Yiheng, Tianle Han, Siyuan Ma, Jiayue Zhang, Yuanyuan Yang, Jiaming Tian, Hao He, Antong Li, Mengshen He, Zhengliang Liu, Zihao Wu, Lin Zhao, Dajiang Zhu, Xiang Li, Ning Qiang, Dingang Shen, Tianming Liu, and Bao Ge. 2023. Summary of chatgpt-related research and perspective towards the future of large language models. *Meta-Radiology*, 1(2):100017.

Liu, Yinhan, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. 2019. Roberta: A robustly optimized bert pretraining approach.

Lotter, Leon D, Simon H Kohl, Christian Gerloff, Laura Bell, Alexandra Niephaus, Jana A Kruppa, Juergen Dukart, Martin Schulte-Rüther, Vanessa Reindl, and Kerstin Konrad. 2023. Revealing the neurobiology underlying interpersonal neural synchronization with multimodal data fusion. *Neuroscience & biobehavioral reviews*, 146:105042.

Maity, Krishanu, Prince Jha, Sriparna Saha, and Pushpak Bhattacharyya. 2022. A multitask framework for sentiment, emotion and sarcasm aware cyberbullying detection from multi-modal code-mixed memes. In *Proceedings of the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 1739–1749.

Maki, Yohko, Tomoharu Yamaguchi, Tatsuya Koeda, and Haruyasu Yamaguchi. 2013. Communicative competence in alzheimer's disease: Metaphor and sarcasm comprehension. *American Journal of Alzheimer's Disease & Other Dementias®*, 28(1):69–74.

Martin, Louis, Benjamin Muller, Pedro Javier Ortiz Suárez, Yoann Dupont, Laurent Romary, Éric de la Clergerie, Djamé Seddah, and Benoît Sagot. 2020. Camembert: a tasty french language model. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, Association for Computational Linguistics.

McFee, Brian, Colin Raffel, Dawen Liang, Daniel Ellis, Matt Mcvicar, Eric Battenberg, and Oriol Nieto. 2015. librosa: Audio and music signal analysis in python.

Michael, Steve and Amalia Zahra. 2019. Automatic sarcasm detection with textual and acoustic data. *International Journal of Recent Technology and Engineering*, 8(4):1357–1360.

Mikolov, Tomas, Kai Chen, Greg Corrado, and Jeffrey Dean. 2013. Efficient estimation of word representations in vector space.

Mishra, Abhijit, Kuntal Dey, and Pushpak Bhattacharyya. 2017. Learning cognitive features from gaze data for sentiment and sarcasm classification using convolutional neural network. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 377–387.

Mishra, Abhijit, Diptesh Kanojia, and Pushpak Bhattacharyya. 2016. Predicting readers' sarcasm understandability by modeling gaze behavior. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 30.

Misra, Rishabh and Prahal Arora. 2023. Sarcasm detection using news headlines dataset. *AI Open*, 4:13–18.

O'Halloran, Kay L, Gautam Pal, and Minhao Jin. 2021. Multimodal approach to analysing big social and news media data. *Discourse, Context & Media*, 40:100467.

Oquab, Maxime, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel Haziza, Francisco Massa, Alaaeldin El-Nouby, Mahmoud Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Hervé Jegou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. 2024. Dinov2: Learning robust visual features without supervision.

Ott, Myle, Sergey Edunov, Alexei Baevski, Angela Fan, Sam Gross, Nathan Ng, David Grangier, and Michael Auli. 2019. fairseq: A fast, extensible toolkit for sequence modeling. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (Demonstrations)*, pages 48–53, Association for Computational Linguistics, Minneapolis, Minnesota.

O'Halloran, Kay L. 2011. Multimodal discourse analysis. *The Bloomsbury handbook of discourse analysis*, pages 249–282.

Panayotov, Vassil, Guoguo Chen, Daniel Povey, and Sanjeev Khudanpur. 2015. Librispeech: An asr corpus based on public domain audio books. In *2015 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 5206–5210.

Parcalabescu, Letitia, Nils Trost, and Anette Frank. 2021. What is multimodality?

Peters, Matthew E., Mark Neumann, Mohit Iyyer, Matt Gardner, Christopher Clark, Kenton Lee, and Luke Zettlemoyer. 2018. Deep contextualized word representations.

Peyré, Gabriel, Marco Cuturi, et al. 2019. Computational optimal transport: With applications to data science. *Foundations and Trends® in Machine Learning*, 11(5-6):355–607.

Poria, Soujanya, Devamanyu Hazarika, Navonil Majumder, Gautam Naik, Erik Cambria, and Rada Mihalcea. 2019. MELD: A multimodal multi-party dataset for emotion recognition in conversations. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 527–536, Association for Computational Linguistics, Florence, Italy.

Pramanick, Shraman, Aniket Roy, and Vishal M. Patel. 2022. Multimodal learning using optimal transport for sarcasm and humor detection. In

Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), pages 3930–3940.

Radford, Alec, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. Learning transferable visual models from natural language supervision.

Rasheed, Hanoona, Muhammad Uzair Khattak, Muhammad Maaz, Salman Khan, and Fahad Shahbaz Khan. 2023. Fine-tuned clip models are efficient video learners.

Ravanelli, Mirco, Jianyuan Zhong, Santiago Pascual, Pawel Swietojanski, Joao Monteiro, Jan Trmal, and Yoshua Bengio. 2020. Multi-task self-supervised learning for robust speech recognition.

Ray, Anupama, Shubham Mishra, Apoorva Nunna, and Pushpak Bhattacharyya. 2022. A multimodal corpus for emotion recognition in sarcasm. *arXiv preprint arXiv:2206.02119*.

Rothermich, Kathrin, Ayotola Ogunlana, and Natalia Jaworska. 2021. Change in humor and sarcasm use based on anxiety and depression symptom severity during the covid-19 pandemic. *Journal of Psychiatric Research*, 140:95–100.

Sanh, Victor, Lysandre Debut, Julien Chaumond, and Thomas Wolf. 2020. Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter.

Schifanella, Rossano, Paloma De Juan, Joel Tetreault, and Liangliang Cao. 2016. Detecting sarcasm in multimodal social platforms. In *Proceedings of the 24th ACM international conference on Multimedia*, pages 1136–1145.

Schneider, Steffen, Alexei Baevski, Ronan Collobert, and Michael Auli. 2019. wav2vec: Unsupervised pre-training for speech recognition.

Sennrich, Rico, Barry Haddow, and Alexandra Birch. 2016. Neural machine translation of rare words with subword units. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1715–1725, Association for Computational Linguistics, Berlin, Germany.

Shamay-Tsoory, Simone G, Rachel Tomer, and Judith Aharon-Peretz. 2005. The neuroanatomical basis of understanding sarcasm and its relationship to social cognition. *Neuropsychology*, 19(3):288.

Stolcke, Andreas, Klaus Ries, Noah Coccaro, Elizabeth Shriberg, Rebecca Bates, Daniel Jurafsky, Paul Taylor, Rachel Martin, Carol Van Ess-Dykema, and Marie Meteer. 2000. Dialogue act modeling for automatic tagging and recognition of conversational speech. *Computational Linguistics*, 26(3):339–374.

Su, Hang, Wen Qi, Jiahao Chen, Chenguang Yang, Juan Sandoval, and Med Amine Laribi. 2023. Recent advancements in multimodal human–robot interaction. *Frontiers in Neurorobotics*, 17.

Tomar, Mohit, Abhisek Tiwari, Tulika Saha, and Sriparna Saha. 2023. Your tone speaks louder than your face! modality order infused multi-modal sarcasm detection. In *Proceedings of the 31st ACM International Conference on Multimedia*, pages 3926–3933.

Tomar, Mohit Singh, Tulika Saha, Abhisek Tiwari, and Sriparna Saha. 2024. Action and reaction go hand in hand! a multi-modal dialogue act aided sarcasm identification. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*, pages 298–309, ELRA and ICCL, Torino, Italia.

Vaswani, Ashish, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. 2023. Attention is all you need.

Wu, Yang, Yanyan Zhao, Xin Lu, Bing Qin, Yin Wu, Jian Sheng, and Jinlong Li. 2021. Modeling incongruity between modalities for multimodal sarcasm detection. *IEEE MultiMedia*, 28(2):86–95.

Xu, Nan, Zhixiong Zeng, and Wenji Mao. 2020. Reasoning with multimodal sarcastic tweets via modeling cross-modality contrast and semantic association. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 3777–3786, Association for Computational Linguistics, Online.

Zhang, Chao, Zichao Yang, Xiaodong He, and Li Deng. 2020. Multimodal intelligence: Representation learning, information fusion, and applications. *IEEE Journal of Selected Topics in Signal Processing*, 14(3):478–493.

Zhang, Yazhou, Jinglin Wang, Yaochen Liu, Lu Rong, Qian Zheng, Dawei Song, Prayag Tiwari, and Jing Qin. 2023. A multitask learning model for multimodal sarcasm, sentiment and emotion recognition in conversations. *Information*

Fusion, 93:282–301.
Zhong, Guoqiang, Li-Na Wang, Xiao Ling, and Junyu Dong. 2016. An overview on data representation learning: From traditional feature learning to recent deep learning. *The Journal of Finance and Data Science*, 2(4):265–278.
Zhu, Yukun, Ryan Kiros, Richard Zemel, Ruslan Salakhutdinov, Raquel Urtasun, Antonio Torralba, and Sanja Fidler. 2015. Aligning books and movies: Towards story-like visual explanations by watching movies and reading books.
Zong, Yongshuo, Oisin Mac Aodha, and Timothy Hospedales. 2023. Self-supervised multimodal learning: A survey.